

Making of Momentum

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Summary

- ▶ The paper develops a framework to quantify the origins of momentum returns using holdings data.
- ▶ This allows distinguishing between/rejecting different theories of momentum.
- ▶ Two great and important contributions:
 1. Add dynamics and preference for momentum to Demand System Approach to Asset Pricing (KY, 2019).
 2. Decompose momentum returns into component from i) arbitrage intensity/term-structure of elasticities and ii) demand shocks.
- ▶ Findings:
 1. Downward-sloping term structure of elasticities: Investors become *less* elastic over time! **Surprising!**
Bogousslavsky et al. [2021] Greenwood [2005], Coval and Stafford [2007], Duffie [2010], Edmans et al. [2012], Lou [2012]
 2. The term structure of elasticities, not persistent demand shocks, is what drives momentum returns.

Comments

1. Separate momentum preference from long-run elasticity.
2. Top down vs bottom up approach.
3. Not necessarily a new theory of momentum: But allows distinguishing between different existing theories.

Comment 1

- ▶ The cross-sectional demand curve in KY (2019):

$$\log w_{i,t}(n) = (1 - \xi_i) \log P_t(n) + \epsilon_{i,t}(n)$$

- ▶ Demand elasticity ξ_i estimated in levels.
- ▶ No time horizon/distinction between long- and short-term elasticities.
- ▶ The demand curve in this paper

$$\log w_{i,t}(n) = (1 - \xi_{\text{Recent},i}) \Delta p_t(n) + (1 - \xi_{\text{Long-Term},i}) \Delta p_{t-s}(n) + \epsilon_{i,t}(n)$$

- ▶ Unclear what $\xi_{\text{Long-Term},i}$ is measuring. Mix of long-term elasticity and momentum preference.
- ▶ Standard way to differentiate between long/short-term elasticities in economics: **Partial Adjustment Model**

$$q_{i,t}(n) = \rho q_{i,t-1}(n) - \underbrace{\xi_{\text{Recent},i} p_t(n)}_{\text{Elasticity}} + \underbrace{\beta_{\text{Mom}} \Delta p_{t-s}(n)}_{\text{Momentum}} + \epsilon_{i,t}(n)$$

- ▶ Long-run elasticity $\xi_{\text{Long-Term},i} = \frac{\xi_{\text{Recent},i}}{1-\rho}$ captured via AR($\cdot$) term ρ
 - ▶ Clear separation between mom. demand β_{Mom} and long-run elasticity
- see e.g. Bachetta et al. 2023

Comment 2

- ▶ Two ways of estimating multipliers/elasticities and their term structure:
 1. Top down approach: Regress prices onto demand shocks.
 - ▶ Index inclusions, flow-induced trading, microstructure regressions etc.
 2. Bottom up approach: Estimate elasticities from holdings, then aggregate up.
 - ▶ KY (2019)
- ▶ The two approaches should yield similar multipliers, both in terms of magnitude and in terms of term-structure.

Comment 2

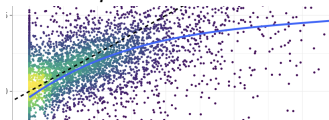
1. FIT Impact Reversal:

Price impact of flows reverts over time (see also: Lou 2012)

	Return $p_t - p_{t-1}$	
	(1)	(2)
$FIT_{t-1-t}(n)$	1.399*** (0.233)	
$FIT_{t-4-t-1}(n)$	-0.354*** (0.095)	

2. Downward Sloping Term Structure:

Price impact of demand shocks increases over time (on average).



- ▶ The theory presented in this paper would imply that the subset of stocks affected by FIT actually have upward-sloping term structures. This is testable!

Comment 3a

- ▶ Framing existing models in terms of demand curves and elasticities that one can estimate is super useful! Allows distinguishing between different theories (see also Haddad et al. [2021])
- ▶ Positive demand coefficient on past price change could be due to preference for momentum or arbitrage intensity.
 - ▶ **Cannot** distinguish between behavioural/risk-based explanation.
 - ▶ But **can** distinguish between two broad categories: Theories based on investors *reacting to past returns* versus *everything else*.
 - ▶ Convince us that this is an important/useful categorization.
- ▶ Different origins of momentum: Can you map your estimation more closely to explicitly **distinguish** between existing explanations?
 - ▶ Often difficult to disentangle demand shock and elasticity.
 - ▶ For example: Elasticity in De Long et al. [1990] $\xi \approx \frac{(1+r)^3}{\gamma\mu\sigma_p^2}$ negatively depends on volatility of non-fundamental demand shock σ_p^2
 - ▶ Similarly, in Vayanos and Woolley [2013] elasticity depends on risk of flow portfolio Σf_p

Comment 3b

- ▶ Conceptually, can we easily disentangle demand elasticities from persistent unexpected demand shocks?
- ▶ Example from paper: No persistence in demand shock $\phi = 1$, upward sloping term structure of elasticities.
 - ▶ $t = 0$: Price equal to fundamental value: 100\$.
 - ▶ $t = 1$: One-time negative demand shock. Prices drop to 95\$ to compensate risk averse short-term arbitrageurs (who are long) with higher expected returns of 5\$.
 - ▶ $t = 2$: Prices move down further to 90\$ because short-term arbitrageurs sell to long-term traders who are more inelastic (require larger compensation to go long).
 - ▶ $t = 3$: Asset pays off 100\$.
- ▶ The short-term arbitrageurs has certain losses of 5\$!

Comment 3b

- ▶ In Vayanos and Woolley [2013] there is an upward-sloping term structure, but only *because* of uncertain non-fundamental demand shocks.
- ▶ Asset with expected payoff: 100\$.
 - ▶ $t = 1$: Investors become aware of potential negative demand shock at $t = 2$. Price either drops to \$80 or stays at \$100 with equal probabilities.
 - ▶ Price only drops to \$95. Why?
 - ▶ Arbitrageur is compensated for “risk” that demand shock does not materialize. **Bird-in-hand effect!**
 - ▶ \$5 with certainty better than \$20 or \$0 with equal probabilities.
 - ▶ $t = 2$: Demand shock realizes and price drops to \$90 on average.
 - ▶ $t = 3$: Asset pays of \$100 (on average).
 - ▶ **Momentum + Reversal**: \$100 $\rightarrow$ \$95 $\rightarrow$ \$90 $\rightarrow$ \$100.
- ▶ In the absence of uncertainty about demand shock, there is no upward sloping term-structure/momentum.

Other Comments

- ▶ Is the trading strategy implementable? You are using the momentum returns in the demand estimation itself. Estimating the term structure over rolling windows (out-of-sample) would be insightful.
- ▶ Nonlinear Price Impact: Selling \$10 Million Shares of Apple depresses price by small amount. Then wait for price to fall and quickly buy back when price dropped?
- ▶ Large micro-structure literature on price impact at higher frequencies: **Concave** impact (“square-root law”) over time and quantities. Seems to be at odds with the **convex** impact found in this paper.
- ▶ Time versus quantity: Is there an invariance for different time-quantity pairs as in Kyle and Obizhaeva [2016]
- ▶ Why is there heterogeneity in the term-structure across stocks? Self-selection of different investors into different habitats? Interplay between demand shock persistence and arbitrage activity?
- ▶ Can you map it more closely to Vayanos and Woolley [2013]?
- ▶ You find that larger stocks are more liquid (higher elasticities)? Isn't this at odds with Haddad et al. [2021]?

Conclusion

- ▶ Super nice paper! Two very important extensions to the demand system:
 1. Dynamics
 2. Incorporate preference for past returns (momentum)
- ▶ Surprising result: Impact of demand shocks are larger at longer horizons.

Comments

1. Estimation: Separate momentum preference from long-run elasticity.
2. Some reduced form tests and more explicit links to existing models could be super helpful.

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